Sukanta Basu

Professor of Empire Innovation
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Education

University of Minnesota — Minneapolis, MN, USA
Ph.D., Civil Engineering — 2000–2004

- Dissertation: "Large-eddy simulation of stably stratified atmospheric boundary layer turbulence: a scale-dependent dynamic modeling approach"

University of Windsor — Windsor, ON, Canada
Master of Applied Science, Environmental Engineering — 1999–2000

Indian Institute of Technology (IIT) — Kharagpur, WB, India
Bachelor of Technology (Honors), Civil Engineering — 1994–1998

Professional Experience

University at Albany — Albany, NY, USA
Professor of Empire Innovation — April, 2023 –
Atmospheric Sciences Research Center

University at Albany — Albany, NY, USA
Full Professor (tenured) — April, 2023 –
Department of Environmental and Sustainable Engineering

Delft University of Technology — Delft, The Netherlands
Associate Professor (permanent position) — September, 2016–March, 2023
Faculty of Civil Engineering and Geosciences

North Carolina State University — Raleigh, NC, USA
Associate Professor (tenured in 2014) — August, 2010–July, 2016
Department of Marine, Earth, and Atmospheric Sciences

Texas Tech University — Lubbock, TX, USA
Assistant Professor — September, 2005–August, 2010
Atmospheric Science Group, Department of Geosciences
Also at Wind Science and Engineering Research Center

University of Minnesota — Minneapolis, MN, USA
Post-doctoral Research Associate — January–June, 2005
St. Anthony Falls Laboratory

Adjunct/Visiting Positions

Università degli Studi di Cagliari — Cagliari, Italy
Visiting Professor — June 1–15, 2019
Dipartimento di Ingegneria Civile, Ambientale e Architettura

North Carolina State University — Raleigh, NC, USA
Adjunct Associate Professor — August, 2016–September, 2018
Department of Marine, Earth, and Atmospheric Sciences

Wageningen University — Wageningen, The Netherlands
Visiting Scientist — May–June, 2008
Meteorology and Air Quality Group

Research Interests

Artificial intelligence and machine learning; atmospheric boundary layer processes; atmospheric optics; numerical weather prediction; renewable energy; and turbulence modeling.

Selected Accomplishments

- Elected Full Member, Sigma Xi, The Scientific Research Honor Society — 2026
- Outstanding Research Award, University at Albany — 2025
  Research & Entrepreneurship Week
- Third place winner of the NASA Airathon: — 2022
  https://www.drivendata.org/competitions/91/competition-air-quality-no2/
- First place winner of the Shell.ai Hackathon: — 2021
  for sustainable and affordable energy; https://www.shell.com/energy-and-innovation/digitalisation/digital-and-ai-competitions/shell-ai-hackathon-for-sustainable-and-affordable-energy.html
- Best teacher awards @ TU-Delft: — 2019
  Geoscience and Remote Sensing track (rank #1)
  Environmental Engineering track (rank #1)
- Winner, Offshore Wind Accelerator wake-modelling challenge — 2019
  Carbon Trust
- Best teacher awards @ TU-Delft: — 2018
  Geoscience and Remote Sensing track (rank #1)
  Environmental Engineering track (rank #2)
- Departmental nominee for the Chancellor's Distinguished Research Award — 2008
  Texas Tech University
- CAREER Award, National Science Foundation (USA) — 2008
- Best Ph.D. dissertation in Civil Engineering — 2005
  awarded by the Department of Civil Engineering, University of Minnesota
- NOAA Climate and Global Change postdoctoral fellowship — 2005
  (declined in order to accept a faculty position at Texas Tech University)
- Doctoral dissertation fellowship — 2003
  awarded by the Graduate School, University of Minnesota

- Sommerfeld fellowship 2000
  awarded by the Department of Civil Engineering, University of Minnesota
- Ontario graduate scholarship awarded by the selection board of 2000
  Ministry of Training, Colleges and Universities, Ontario, Canada
  (declined in order to accept the Sommerfeld fellowship at the University of Minnesota)
- Best final year project in Civil Engineering 1998
  awarded by Indian Institute of Technology, Kharagpur, India

Machine Learning Activities

Selected Participation in Competitions

- *Shell.ai Hackathon for sustainable and affordable energy: Fuel blend properties prediction challenge*
  Selected for level 2 (one of 25 participants out of ∼7000 total),
  https://www.hackerearth.com/challenges/new/competitive/shellai-hackathon-2025/ 2025
- *Solafune: aerosol optical depth (AOD) estimation*
  Ranked 25th out of 307, bronze star
  https://solafune.com/competitions/ca6ee401-eba9-4f7d-95e6-d1b378a17200?menu=about&tab=/ 2024
- *IEEE: hybrid energy forecasting and trading competition*
  Ranked 6th overall,
  https://ieee-dataport.org/competitions/hybrid-energy-forecasting-and-trading-competition 2024
- *Kaggle: LEAP – atmospheric physics using AI (ClimSim)*
  https://www.kaggle.com/competitions/leap-atmospheric-physics-ai-climsim 2024
- *Zindi: AI/ML for 5G energy consumption modeling*
  https://zindi.africa/competitions/aiml-for-5g-energy-consumption-modelling 2023
- *HEROx: American-made challenges – net load forecasting prize*
  Ranked 9th overall, ranked 1st for Amity, OR. https://www.herox.com/net-load-forecasting 2023
- *CVPR: atmospheric turbulence mitigation*
  https://cvpr2023.ug2challenge.org/track2.html 2023
- *NASA Airathon: predict air quality (trace gas track)*
  Ranked 3rd, https://www.drivendata.org/competitions/91/competition-air-quality-no2/ 2022
- *NASA Airathon: predict air quality (particulate-matter, PM2.5 track)*
  Ranked 6th, https://www.drivendata.org/competitions/88/competition-air-quality-pm// 2022
- *Shell.ai Hackathon for sustainable and affordable energy: short-term forecasting of solar radiation*
  Ranked 1st, https://www.shell.com/energy-and-innovation/digitalisation/digital-and-ai-competitions/shell-ai-hackathon-for-sustainable-and-affordable-energy.html 2021
- *Day-ahead electricity demand forecasting: post-covid paradigm.*
  https://dx.doi.org/10.21227/67vy-bs34 2021
- *Drivendata competition on tropical storm intensity prediction*
  Ranked 32nd out of 733.
  https://www.drivendata.org/competitions/72/predict-wind-speeds/leaderboard/ 2020
- *European Electricity Market (EEM) forecasting competition*
  Team leader, Ranked 3rd. 2020

Online Courses

- *Coursera: Sequence models*
  https://www.coursera.org/account/accomplishments/certificate/MD6RLSH6YNX3 2022
- *Coursera: Convolutional neural networks*
  https://www.coursera.org/account/accomplishments/certificate/TKRW88JJDPU2 2022
- *Coursera: Structuring machine learning projects*
  https://www.coursera.org/account/accomplishments/certificate/9VZQC7U2DVZE 2021
- *Coursera: Improving deep neural networks*
  https://www.coursera.org/account/accomplishments/certificate/H6JWL762U88B 2021
- *Coursera: Neural networks and deep learning*
  https://www.coursera.org/account/accomplishments/certificate/HWQTB23HRWNH 2021

Professional Activities

Associate Editor

- *Journal of Geophysical Research – Atmosphere* 2023–Present
- *Wind Energy Science* 2022–2026

Editorial Board Memberships

- *Boundary-Layer Meteorology* 2023–Present
- *Wind Energy* 2019–Present
- *Environmental Fluid Mechanics* 2017–Present
- *Frontiers in Built Environment* 2016–Present
- *Atmosphere* 2020–2021
- *Energies* 2015–2021
- *The Open Atmospheric Science Journal* 2007–2017

Guest Editing

- Special issue on "Recent Advances in Wind Power Meteorology", *Energies* 2020–2021
- Special issue on "Large-Eddy Simulations (LES) of Atmospheric Boundary Layer Flows", *Atmosphere* 2018

Committees & Review Panels

- Member, NATO SET-ET-140 group 2025–Present
  (Atmospheric effects on electro-optical systems)
- Member, IEA Task-51 (https://iea-wind.org/task51/) 2023–Present
  (Forecasting for the weather-driven energy system)
- Member, NATO SET-304 group 2021–2024
  (Modelling, measuring, and mitigating optical turbulence)
- Founding member and coordinator of a new M.Sc. track 2020–2022
  Atmospheric Environment Engineering at TU-Delft
- Chair, Atmospheric Sciences Review Panel, 2021
  Academy of Finland, Helsinki

- Coordinator, met-ocean program line, GROW (https://www.grow-offshorewind.nl/) 2018–2020
- Member, NATO SET-ET-118 group (Modelling, measuring, and mitigating optical turbulence) 2019–2020
- Member, Atmospheric Sciences Review Panel, Academy of Finland, Helsinki 2019, 2020
- Contributing author of "The Netherlands' Long-Term Offshore Wind R&D Agenda" (https://www.topsectorenergie.nl/nieuws/netherlands-long-term-offshore-wind-rd-agenda) 2019
- Member, Site Visit Panel for Strategic Partnership Grants, NSERC, Canada 2019
- Member, CAREER Panel, Energy for Sustainability Program, NSF, USA 2011
- Member, Wind/Wave Energy Panel, Energy for Sustainability Program, NSF, USA 2011
- Member, CAREER Panel on Biofuels and Wind Energy, Energy for Sustainability Program, NSF, USA 2010
- Member, Boundary Layers and Turbulence Committee, American Meteorological Society, USA 2009–2012
- Member, Site Visit Panel Member, Wind Program Evaluation, Arizona Science Foundation, USA 2008
- Member, Committee Member, Lone Star Emmy Educational Foundation Harold Taft Scholarship, USA 2008

Conference & Workshop Organization

- Member of the Organizing Committee, NCAR Summer Colloquium on the Atmospheric Boundary Layer, Boulder, CO, USA (https://edec.ucar.edu/advanced-study-program/asp-colloquia) 2023
- Theme Lead, Wind and Wind Farms, Torque 2022, Delft, the Netherlands (https://www.torque2022.eu/) 2022
- Program Chair, Propagation Through and Characterization of Atmospheric and Oceanic Phenomena, OSA 2019–2021
- Conference Committee Member, Laser Communication and Propagation through the Atmosphere and Oceans, SPIE 2015–2022
- Conference Committee Member, Environmental Effects on Light Propagation and Adaptive Systems, SPIE Remote Sensing 2019–2022
- Theme Lead, Wind and Wind Farms, Torque 2020, Delft, the Netherlands (http://torque2020.org/) 2020
- Scientific Organization Committee, workshop on 'Non-Kolmogorov Turbulence and Associated Phenomena', Fraunhofer IOSB, Ettlingen, Germany 2019
- Conference Committee Member, Propagation Through and Characterization of Atmospheric and Oceanic Phenomena, OSA, Orlando, FL, USA 2018
- Organizing Committee Member, Workshop on the Atmospheric Stable Boundary Layer, Delft, the Netherlands 2017

- Symposium Planning Committee Member, 'Evapotranspiration: Challenges in Measurement and Modeling from Leaf to the Landscape Scale and Beyond', Raleigh, NC, USA — 2014
- Scientific Committee Member, 13th International Conference on Wind Engineering Amsterdam, The Netherlands — 2011
- Co-convener of 'Understanding of Land-Atmosphere Interactions with Models and Observations', AGU Fall Meeting, San Francisco, CA, USA — 2009
- Co-convener of 'Wind Power Meteorology', AGU Fall Meeting, San Francisco, CA, USA — 2008–2010
- Co-convener of 'Atmospheric Turbulence Scaling Mechanisms and their Meteorological Effects', AGU Joint Assembly, Acapulco, Mexico — 2008–2010

Conference Session Chairs, Panel Members

- Panel Member, VAIBHAV Summit (https://innovate.mygov.in/vaibhav-summit/), Government of India — 2020
- Session Chair, Propagation Through and Characterization of Atmospheric and Oceanic Phenomena, OSA — 2020
- Session Chair, Propagation Through and Characterization of Atmospheric and Oceanic Phenomena, OSA, Munich, Germany — 2019
- Session Chair, Propagation Through and Characterization of Atmospheric and Oceanic Phenomena, OSA, Orlando, FL, USA — 2018
- Session Chair, American Meteorological Society (AMS) 21st Symposium on Boundary Layers and Turbulence, Salt Lake City, UT, USA — 2016
- Session Chair, American Meteorological Society (AMS) 20th Symposium on Boundary Layers and Turbulence, Boston, MA, USA — 2012
- Chair, Student Award Committee, American Meteorological Society (AMS) 19th Symposium on Boundary Layers and Turbulence, Keystone, CO, USA — 2010
- Session Chair, "ABL and Turbulence Models for CWE II", the Fifth International Symposium on Computational Wind Engineering, Chapel Hill, NC, USA — 2010
- Discussion Panel Member, "Development, Validation, and Application of Atmospheric Boundary Layer Models and Turbulence Models for CWE", the Fifth International Symposium on Computational Wind Engineering, Chapel Hill, NC, USA — 2010

Miscellaneous Activities

- Winner of the Offshore Wind Accelerator (OWA) wake modelling challenge — 2019
- Chair of the 3rd GEWEX Atmospheric Boundary Layer Study (GABLS) Large-Eddy Simulation Intercomparison — 2008–2010

Reviewing Activities

Please visit https://publons.com/researcher/2777086/sukanta-basu/ for further information.

Papers reviewed for: Advances in Meteorology; ASME wind energy symposium; Applied Optics; Atmosphere; Boundary-Layer Meteorology; Energies; Environmental Fluid Mechanics; Environmental Research Letters; Euro

Physics Letters; Geophysical Research Letters; Irrigation Science; Journal of Applied Meteorology and Climatology; Journal of Applied Physics; Journal of Applied Remote Sensing; Journal of Climate; Journal of Fluid Mechanics; Journal of Geophysical Research; Journal of the Atmospheric Sciences; Journal of Hydrometeorology; Journal of Renewable and Sustainable Energy; Journal of the Optical Society of America B; Journal of the Meteorological Society of Japan; Journal of Turbulence; Journal of Wind Engineering and Industrial Aerodynamics; Meteorology and Atmospheric Physics; Monthly Weather Review; Nonlinear Processes in Geophysics; Optics Express; Optics Letters; Physica D; Physics Letters A; Physics of Fluids; Quarterly Journal of Royal Meteorological Society; Tellus B; The European Physical Journal B; The Open Atmospheric Science Journal; Transactions of the ASABE; Water Resources Research; Weather and Forecasting; Wind Energy; Wind Energy Science.

Book chapters reviewed for: AMS monograph series (100 years of progress in Boundary-layer Meteorology).

Proposals reviewed for: National Science Foundation, USA; Academy of Finland; ETH Zurich, Switzerland; Natural Environment Research Council, UK; The Natural Sciences and Engineering Research Council of Canada; Arizona Science Foundation; Innovation and Technology Commission, The Government of Hong Kong; the Netherlands Organisation for Scientific Research (NWO); Technology Foundation STW, the Netherlands; Foundation for Fundamental Research on Matter (FOM), the Netherlands.

External PhD dissertation examiner: ForWind - Center for Wind Energy Research, Carl von Ossietzky Universität, Oldenburg, Germany; Urban and Coastal Atmosphere Dynamics Group, Ecole Centrale Nantes, France; Physics and Material Sciences, ONERA, Aix Marseille University, France; Department of Geography, KU Leuven, Belgium; Civil Engineering, The University of Sydney, Australia; Department of Civil, Architectural, and Environmental Engineering, The University of Texas-Austin; School of Mechanical & Aerospace Engineering, Nanyang Technological University, Singapore; Thermal and Fluid Engineering, Indian Institute of Technology–BHU.

External MSc thesis examiner: Technical University of Denmark (European Wind Energy masters program).

External course evaluator: Atmospheric Modelling course (MAQ 31806), Meteorology and Air Quality Group, Wageningen University, The Netherlands.

Sponsored Research

- Funding Agency: National Aeronautics and Space Administration (NASA) 2026–2028
  Project Title: Coupling NASA Earth observation and weather-climate foundation models for power outage management
  PI: Sukanta Basu
  Funds: $302K (Basu's group); total: $500K
- Funding Agency: NSF-IUCRC (WISER) 2026–2027
  Project Title: North American outage prediction model
  Co-PI: Sukanta Basu (one of several)
  Funds: $37,500 (Basu's group); total: $225,000
- Funding Agency: Department of Financial Services, New York State 2025–2028
  Project Title: Assessing climate exposure and financial impacts at the census tract level in New York state
  PI: Sukanta Basu
  Co-PIs: Fangqun Yu, Jan Woodcock
  Funds: $1,611,585

- Funding Agency: NSF-IUCRC (WISER) 2025–2026
  Project Title: Deep learning-based nowcasting of damaging winds (Phase II)
  PI: Sukanta Basu
  Funds: $50,000
- Funding Agency: Optonica, LLC 2024–2025
  (Prime: The Defense Advanced Research Projects Agency (DARPA),
  Joint Directed Energy Transition Office)
  Project Title: Atmospheric propagation basic research
  PI: Sukanta Basu
  Funds: $89,418
- Funding Agency: NSF-IUCRC (WISER) 2024–2025
  Project Title: Deep learning-based nowcasting of damaging winds (Phase I)
  PI: Sukanta Basu
  Funds: $50,000
- Funding Agency: University of Dayton Research Institute
  (Prime: US Air Force Research Laboratory, Joint Directed Energy Transition Office) 2023–2024
  Project Title: University at Albany on SOARING OTTER
  PI: Sukanta Basu
  Funds: $93,495
- Funding Agency: Horizon 2020, European Green Deal 2021–2025
  Project Title: EUropean - Scalable and Complementary Offshore Renewable Energy Sources
  Lead organization: Dutch Marine Energy Centre (DMEC);
  co-PIs from TU-Delft: George Lavidas and Sukanta Basu
  Funds: €448,000 (Basu's group); total: €34M
- Funding Agency: Dutch Research Council (NWO) 2021–2026
  Project Title: Optical wireless super highways
  Lead organization: TU-Delft;
  PI: Eberhard Gill; co-PI: Sukanta Basu (one of several)
  Funds: €250,000 (Basu's group); total: €5.1M
- Funding Agency: TU Delft Institute for Computational Science and Engineering 2021–2023
  Project Title: Reliable estimation of extreme wind gusts over the North Sea:
  A convolutional neural network-based framework
  PI: Sukanta Basu; co-PI: Simon Watson
  Funds: €80,000
- Funding Agency: TKI, Wind op Zee, The Netherlands 2020–2022
  Project Title: WINS50: Winds of the North Sea in 2050
  Lead organization: Whiffle, BV; co-PI: Sukanta Basu and others
  Funds: €462,869 (TU-Delft part)
- Funding Agency: CENER, Spain 2020
  Project Title: Offshore wind accelerator: wake modelling challenge
  PI: Sukanta Basu
  Funds: €9,640
- Funding Agency: Sports Engineering Institute @ TU-Delft 2020
  Project Title: Tokyo innovation funds: atmospheric modelling
  in support of the Sail Ghost project
  PI: Sukanta Basu, co-PI: Anoek Van Vlaardingen
  Funds: €25,000

- Funding Agency: Carbon Trust, UK — 2017–2018
  Project Title: Offshore wind accelerator: boundary layer profiling
  PI: Sukanta Basu, co-PI: Simon Watson
  Funds: £44,680
- Funding Agency: Sailing Innovation Center, The Netherlands — 2017
  Project Title: Sail ghost of Enoshima
  PI: Sukanta Basu
  Funds: €9,075
- Funding Agency: NSF (AGS-EAGER), USA — 2016–2018
  Project Title: Identifying the limitations of the contemporary planetary boundary layer schemes using an extended self-similarity-based framework
  PI: Sukanta Basu
  Funds: $63,189
- Funding Agency: NSF (CBET), USA — 2013–2017
  Project Title: A retrospective assessment and future projection of thunderstorm impacts on the field performance of wind turbines
  PI: Lance Manuel (University of Texas-Austin), co-PI: Sukanta Basu
  Funds: $199,810 (NCSU-part)
- Funding Agency: Department of Defense, USA — 2012–2016
  Project Title: Wave optics of deep atmospheric turbulence: from underlying physics towards predictive modeling, mitigation and exploitation
  PI: Mikhail Vorontsov (University of Dayton), co-PI: Sukanta Basu
  Funds: $500,000 (NCSU-part)
- Funding Agency: Renaissance Computing Institute (RENCI), USA — 2011–2014
  Project Title: Micro-siting of wind turbines over complex terrains utilizing the OpenFOAM CFD toolbox and the WRF model
  PI: Sukanta Basu
  Funds: $82,757
- Funding Agency: National Renewable Energy Lab, USA — 2011–2012
  Project Title: Modeling the stable atmospheric boundary layer with computational fluid dynamics for wind energy application
  PI: Sukanta Basu
  Funds: $4,272
- Funding Agency: NSF (CBET), USA — 2010–2014
  Project Title: On turbine loads assessment for ultimate and fatigue limit states for different atmospheric boundary layer stability conditions
  PI: Lance Manuel (University of Texas-Austin), co-PI: Sukanta Basu
  Funds: $123,140 (NCSU-part)
- Funding Agency: Department of Energy, USA — 2010–2013
  Project Title: Enhancing short term wind energy forecasting for improved utility operations
  PI: Lead organization: AWS Truepower, co-PI: Sukanta Basu
  Funds: $75,631 (NCSU-part)
- Funding Agency: NSF (CAREER award), USA — 2008–2014
  Project Title: Towards better representation of the nocturnal low-level jets in new generation large-eddy and mesoscale models

PI: Sukanta Basu
Funds: $505,060

- Funding Agency: Norman Hackerman Advanced Research, USA 2008–2010
  Project Title: Atmospheric stability considerations in design of wind turbines against fatigue
  PI: Lance Manuel (University of Texas-Austin), co-PI: Sukanta Basu
  Funds: $51,776 (TTU-part)
- Funding Agency: NSF (Polar science program), USA 2006–2010
  Project Title: Understanding, parameterizing and modeling the strongly stratified atmospheric boundary layer processes over the Antarctic Plateau
  PI: Sukanta Basu
  Funds: $173,000
- Funding Agency: Texas Advance Research Program, USA 2006–2009
  Project Title: Characterization and simulation of turbulence in stably stratified atmospheric boundary layers
  PI: Sukanta Basu, co-PI: Xiaoning Gilliam
  Funds: $86,000
- Funding Agency: Texas Tech University, USA 2006–2009
  Project Title: Innovative technologies to investigate fine-scale atmospheric motions and their impact
  PI: John Schroeder, co-PI: Sukanta Basu and others
  Funds: $1,000,000

Key Collaborators

Steve Fiorino (Air Force Institute of Technology, USA); Bert Holtslag (Wageningen University, The Netherlands); Lance Manuel (University of Texas at Austin, USA); Mikhail Vorontsov (University of Dayton, USA); Simon Watson (TU-Delft, The Netherlands).

Journal Publications

Google Scholar Citations: https://scholar.google.com/citations?user=08bv9p8AAAAJ&hl=en
Researchgate Page: http://www.researchgate.net/profile/Sukanta_Basu
ORCID: 0000-0002-0507-5349
SCOPUS: 7403656752
ResearcherID: F-9286-2011

Graduate students and post-docs of my research group are indicated by * and †, respectively.

[83] Baki†, H., & **Basu**, **S.** (2026). A Chebyshev polynomial-based wind speed profile characterization framework: Applications in mesoscale model evaluation. *Wind Energy*, *29*, e70080.

[82] Baki†, H., Pierzyna*, M., & **Basu**, **S.** (2026). S2G deep interpolation: A deep learning framework for generating high-resolution wind gust fields from sparse Mesonet observations. *Artificial Intelligence for the Earth Systems*, *5*(4). https://doi.org/10.1175/AIES-D-25-0111.1

[81] **Basu**, **S.** (2026). Leveraging deep learning-based foundation models for optical turbulence (C_n^2) estimation under data scarcity [**Editor's Pick**]. *Applied Optics*, *65*, 2497–2504.

[80] Kosović, B., **Basu**, **S.**, Berg, J., Berg, L. K., Haupt, S. E., Larsén, X. G., Peinke, J., Stevens, R. J. A. M., Veers, P., & Watson, S. (2026). Impact of atmospheric turbulence on performance and loads of wind turbines: Knowledge gaps and research challenges. *Wind Energy Science*, *11*, 509–555.

[79] Pierzyna*, M., **Basu**, **S.**, & Saathof, R. (2026). OTProf: Estimating high-resolution profiles of optical turbulence (C_n^2) from reanalysis using deep learning. *Artificial Intelligence for the Earth Systems*, under review, arXiv:2604.09346.

[78] Baki†, H., **Basu**, **S.**, & Lavidas, G. L. (2025). Modelling frontal low-level jets and associated extreme wind power ramps over the North Sea. *Wind Energy Science*, *10*, 1575–1609.

[77] Pierzyna*, M., **Basu**, **S.**, & Saathof, R. (2025). OTCliM: Generating a near-surface climatology of optical turbulence strength (C_n^2) using gradient boosting. *Artificial Intelligence for the Earth Systems*, *4*, e240076.

[76] Vorontsov, M., Polnau, E., Hammel, S., **Basu**, **S.**, & Hettiarachchi, D. L. N. (2025). Analysis of atmospheric turbulence dynamics during the total solar eclipse with AI-based sensing. *Journal of the Optical Society of America A*, *42*, 1112–1119.

[75] Baki†, H., & **Basu**, **S.** (2024). Estimating high-resolution profiles of wind speeds from a global reanalysis dataset using TabNet. *Environmental Data Science*, *3: e32*, 1–10.

[74] Baki†, H., **Basu**, **S.**, & Lavidas, G. L. (2024). Estimating the offshore wind power potential of Portugal by utilizing gray-zone atmospheric modeling. *Journal of Renewable and Sustainable Energy*, *16*, 063306.

[73] Pierzyna*, M., Hartogensis, O., **Basu**, **S.**, & Saathof, R. (2024). Intercomparison of flux, gradient, and variance-based optical turbulence (C_n^2) parameterizations. *Applied Optics*, *63*, E107–E119.

[72] Annis, S., Badas, M. G., Horváth, A., & **Basu**, **S.** (2023). Quantifying the impacts of the angle of attack on the morphology of atmospheric von Kármán vortex streets. *J. Geophysical Research – A*, *128*, e2023JD039125.

[71] Cheneka*, B. R., Watson, S. J., & **Basu**, **S.** (2023). Quantifying the impacts of synoptic weather patterns on North Sea wind power production and ramp events under a changing climate. *Energy and Climate Change*, *4*, 100113.

[70] Kartal†, S., **Basu**, **S.**, & Watson, S. J. (2023). A decision tree-based measure-correlate-predict approach for peak wind gust estimation from a global reanalysis dataset. *Wind Energy Science*, *8*, 1533–1551.

[69] Pierzyna*, M., Saathof, R., & **Basu**, **S.** (2023). Π-ML: A dimensional analysis-based machine learning parameterization of optical turbulence in the atmospheric surface layer [**Editor's Pick**]. *Optics Letters*, *48*, 4484–4487.

[68] **Basu**, **S.**, & Holtslag, A. A. M. (2022a). A novel approach for deriving the stable boundary layer height and eddy viscosity profiles from the Ekman equations. *Boundary-Layer Meteorology*, *187*, 105–115.

[67] **Basu**, **S.**, & Holtslag, A. A. M. (2022b). Revisiting and revising Tatarskii's formulation for the temperature structure parameter (C_T^2) in atmospheric flows. *Environmental Fluid Mechanics*, *22*, 1107–1119.

[66] Li*, B., **Basu**, **S.**, & Watson, S. J. (2022). Automated identification of "Dunkelflaute" events: A convolutional neural network-based autoencoder approach. *Artificial Intelligence for the Earth Systems*, *1: e220015*, 1–17.

[65] Veers, P., Dykes, K., **Basu**, **S.**, Bianchini, A., Clifton, A., Green, P., Holttinen, H., Kitzing, L., Kosovic, B., Lundquist, J. K., Meyers, J., O'Malley, M., Shaw, W. J., & Straw, B. (2022). Grand challenges: Wind energy research needs for a global energy transition. *Wind Energy Science*, *7*, 2491–2496.

[64] **Basu**, **S.**, DeMarco*, A. W., & He†, P. (2021). On the dissipation rate of temperature fluctuations in stably stratified flows. *Environmental Fluid Mechanics*, *21*, 63–82.

[63] **Basu**, **S.**, He†, P., & DeMarco*, A. W. (2021). Parameterizing the energy dissipation rate in stably stratified flows. *Boundary-Layer Meteorology*, *178*, 167–184.

[62] **Basu**, **S.**, & Holtslag, A. A. M. (2021). Turbulent Prandtl number and characteristic length scales in stably stratified flows: Steady-state analytical solutions. *Environmental Fluid Mechanics*, *21*, 1273–1302.

[61] Cheneka*, B. R., Watson, S. J., & **Basu**, **S.** (2021). Associating synoptic-scale weather patterns with aggregated offshore wind power production and ramps. *Energies*, *14*, 3903.

[60] Li*, B., **Basu**, **S.**, Watson, S. J., & Russchenberg, H. W. J. (2021a). A brief climatology of Dunkelflaute events over and surrounding the North and Baltic sea areas. *Energies*, *14*, 6508.

[59] Li*, B., **Basu**, **S.**, Watson, S. J., & Russchenberg, H. W. J. (2021b). Mesoscale modeling of a 'Dunkelflaute' event. *Wind Energy*, *24*, 5–23.

[58] Lu, N.-Y., Manuel, L., Hawbecker*, P. H., & **Basu**, **S.** (2021). A simulation study on risks to wind turbine arrays from thunderstorm downbursts in different atmospheric stability conditions. *Energies*, *14*, 5407.

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Book Chapters

Graduate students and post-docs of my research group are indicated by * and †, respectively.

[2] Basu, S. (2022). Vertical wind speed profiles in atmospheric boundary layer flows, Wind Energy Engineering, 2nd Edition, edited by T. Letcher, Elsevier. ISBN: 9780323993531.

[1] Basu, S. (2012). Turbulent flow modeling, Handbook of Environmental Fluid Dynamics, Volume Two, edited by H. J. S. Fernando, CRC Press/Taylor & Francis Group, LLC. ISBN: 978-1-4665-5601-0.

Conference Proceedings

Graduate students and post-docs of my research group are indicated by * and †, respectively.

[17] Roy Sarkar, D., Basu, S., Manuel, L., and Goswami, S. (2025). Uncertainty-aware optimization in engineered systems via gradient boosting and differential evolution, NASA and DNV UQ Challenge, in Proceedings of the 33rd European Safety and Reliability & 33rd Society for Risk Analysis Europe Conference, https://rpsonline.com.sg/proceedings/esrel-sra-e2025/html/toc.html

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[10] Basu, S., and He†, P. (2015). Estimating refractive index structure parameter (C_n^2) profiles in the atmosphere: A wavelet transform-based approach, Proceedings of SPIE Optics & Photonics, doi: 10.1117/12.2188195.

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[3] Basu, S., Holtslag, A. A. M., and Bosveld, F. C. (2011). GABLS3-LES intercomparison study, Workshop Proceedings – ECMWF Workshop on Diurnal Cycles and the Stable Boundary Layer, 7-10 November, Reading, UK, pages 75-82.

[2] Holtslag, A. A. M., Svensson, G., Basu, S., Beare, R. J., Bosveld, F. C., and Cuxart, J. (2011). Overview of the GEWEX Atmospheric Boundary Layer Study (GABLS), Workshop Proceedings – ECMWF Workshop on Diurnal Cycles and the Stable Boundary Layer, 7-10 November, Reading, UK, pages 11-24.

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Invited Presentations

Presenters are indicated by asterisks.

[29] Basu, S. (2026). "AIML-driven modeling of atmospheric turbulence for weather, energy, and defense applications", 6 March, Johns Hopkins University, Baltimore, USA.

[28] Basu, S. (2025). "AIML-driven modeling of atmospheric turbulence for weather, energy, and defense applications", 19 November, University of Alabama, Huntsville, USA.

[27] Basu, S. (2022). "Outer length scales in nocturnal stable boundary layers", 11-15 July, Imaging & Applied Optics Congress, Vancouver, Canada.

[26] Basu, S. (2022). "Optical turbulence modelling", Keynote lecture, 7-10 June, International conference on Target and Background Modeling & Simulation, Bagnères-de-Bigorre, France.

[25] Basu, S. (2019). "Wind energy developments in a changing climate: how can mesoscale modelling help?", 23-25 July, School of Excellence, University of Messina, Italy.

[24] Basu, S. (2019). "WRF modelling for wind energy applications", 10-14 June, Università degli Studi di Cagliari, Italy.

[23] Basu, S. (2019). "Mesoscale modeling of atmospheric flows for wind energy applications", 6-10 May, von Kármán Institute for Fluid Dynamics, Belgium.

[22] Basu, S. (2017). "Generating realistic inflow conditions for next-generation wind turbines", 13 November, Wageningen University, the Netherlands.

[21] Basu, S. (2017). "Idealized and realistic large-eddy simulations of optical turbulence (C_n^2)", 26-29 June, Propagation through and Characterization of Atmospheric and Oceanic Phenomena (pcAOP), San Francisco, CA, USA.

[20] Basu, S. (2017). "Multiscale modeling of atmospheric refraction and turbulence", 1 June, Trends and Developments in Laser Based Dimensional Metrology, Lorentz Center, Leiden, The Netherlands.

[19] Basu, S. (2017). "Mesoscale modeling of atmospheric flows for wind energy applications", 28 March, von Kármán Institute for Fluid Dynamics, Belgium.

[18] Basu, S. (2016). "On the scaling-based estimation of optical turbulence in the atmosphere", 27-29 June, Propagation through and Characterization of Atmospheric and Oceanic Phenomena (pcAOP), Washington DC, USA.

[17] Basu, S. (2016). "Quasi-universal scaling of boundary-layer wind speeds in mesoscale regime", University of Oklahoma, Norman, OK, USA.

[16] Basu*, S., and He, P. (2015). "Extracting various similarity formulations from an extensive database of direct and large-eddy simulations of stably stratified flows", 27 July, US National Congress on Computational Mechanics, San Diego, CA, USA.

[15] Basu*, S., and He, P. (2015). "Estimating optical turbulence in the atmosphere utilizing the inherent vertical scaling characteristics of temperature fields", 7-11 June, Propagation through and Characterization of Distributed Volume Turbulence (pcDVT), Washington DC, USA.

[14] Basu, S. (2015). "Mesoscale modeling of atmospheric flows for wind energy applications", 25 February, von Kármán Institute for Fluid Dynamics, Belgium.

[13] Basu, S. (2014). "Multiscale modeling of atmospheric refraction and turbulence", 5 November, NATO ET-87 Study Group Meeting, University of Dayton, Dayton, OH, USA.

[12] Basu*, S., and He, P. (2014). "Quantifying the dependence of temperature and refractive index structure parameters on atmospheric stability using direct and large-eddy simulations", 13-17 July, Propagation through and Characterization of Distributed Volume Turbulence (pcDVT), Seattle, WA, USA.

[11] Basu, S. (2013). "Can the LES-Generated datasets complement field measurements?", 20 November,

Brazilian Micrometeorology Workshop, Santa Maria, Brazil.

[10] Basu, S. (2013). "Addressing a few problems in stable boundary layer turbulence", 10 April, Environmental Protection Agency, Raleigh, NC, USA.

[9] Basu, S. (2013). "Mesoscale modeling of atmospheric flows for wind energy applications", 13 March, von Kármán Institute for Fluid Dynamics, Belgium.

[8] Basu, S. (2012). "Addressing a few problems in stable boundary layer turbulence", 20 June, National Center for Atmospheric Research, Boulder, CO, USA.

[7] Basu, S. (2011). "GABLS3 LES Intercomparison Study", ECMWF/GABLS Workshop on Diurnal cycles and the stable atmospheric boundary layer, 7-10 November, Reading, UK.

[6] Basu, S. (2010). "Addressing a few emergent challenges in wind power meteorology", Department of Mechanical Engineering and Engineering Science, University of North Carolina-Charlotte, NC, USA.

[5] Basu, S. (2009). "Addressing a few emergent challenges in wind power meteorology", Visher Lecture, Department of Geography, University of Indiana, Bloomington, IN, USA.

[4] Basu*, S., Ruiz-Columbié, A., and Harshan, S. (2008). "Deriving Monin-Obukhov similarity functions from dynamic large-eddy simulations", American Geophysical Union Fall Meeting, 15-19 December, San Francisco, CA, USA.

[3] Basu, S. (2007). "NEW LES intercomparison case." Presentation at the GABLS Workshop, Stockholm, Sweden.

[2] Basu, S. (2007). "Looking beyond dynamic Smagorinsky models for atmospheric boundary layer simulations." Presentation at the Department of Atmospheric Sciences, University of Washington, Seattle, WA, USA.

[1] Basu, S. (2004). "Understanding stably stratified atmospheric boundary-layer turbulence: integration of statistical and dynamical approaches with large-eddy simulations." Presentation at the International Center for Theoretical Physics, 24 April, Trieste, Italy.

Teaching

- **University at Albany** 2024–Present
  - *ESE 332: Air quality modeling (Spring 2024-2026)*
- **Delft University of Technology** 2016–2023
  - *CIE 4605: Atmospheric Turbulence (Quarter 3, 2017–2022)*
  - *CIE 4707: Air Quality (Quarter 3, 2017–2022)*
  - *AESB 2120: Signal Processing with MATLAB (Quarter 1, 2017–2018)*
  - *AESB 2121: Signals and Systems with PYTHON (Quarter 1, 2019–2022)*
- **North Carolina State University** 2010–2016
  - *MEA 455: Micrometeorology (Fall 2011, 2012, 2014)*
  - *MEA 463: Fluid Physics (Fall 2014)*
  - *MEA 510: Air Pollution Meteorology (Fall 2013, 2015)*
  - *MEA 593: Wind Power Meteorology (Spring 2012, 2014, 2016)*
  - *MEA 707: Planetary Boundary Layer (Spring 2011, 2013, 2015)*
- **Texas Tech University** 2005–2010
  - *ATMO 1300: Introduction to Atmospheric Science (Fall 2005–2008, Spring 2009)*
  - *ATMO 5301: Wind Power Meteorology (Spring 2009–2010)*
  - *ATMO 5301: Atmospheric Dispersion Modeling (Spring 2006, 2008)*

– *ATMO 5319: Boundary-layer Meteorology (Fall 2007, 2009)*
– *ATMO 5332: Regional-scale Numerical Weather prediction (Spring 2007)*

Research Group

Current Research Scientist

- Dr. Chau Lam (Chris) Yu

Current Post-doctoral Research Associates

- Dr. Harish Baki

Current Graduate Advisees

- Ekaterina Belash (Ph.D., UAlbany) – chair: Sukanta Basu
- Emmanuel Kipchirchir (M.S., UAlbany) – chair: Sukanta Basu
- Kazimir Nyzio (Appl. M.S., UAlbany) – mentor: Sukanta Basu

Former Post-doctoral Research Associates

- Dr. Jaya Singh (University at Albany)
  Current position: Postdoctoral Researcher, IIT-BHU, India
- Dr. Vikas Dwivedi (University at Albany)
  Current position: Postdoctoral Researcher, INSA, Lyon, France
- Dr. Serkan Kartal (TU-Delft)
  Current position: Assistant Professor, Computer Engineering, Cukurova University, Adana, Turkey
- Dr. Ping He (NCSU)
  Current position: Assistant Professor, Department of Aerospace Engineering,
  Iowa State University, Ames, USA
- Dr. Arquímedes Ruiz Columbié (TTU)
  Current position: Lecturer, Texas Wind Energy Institute,
  Texas Tech University, Lubbock, USA

Former Ph.D. Advisees

- Maximilian Pierzyna (TU-Delft; co-advised with Rudolf Saathof) 2026
  Dissertation: Machine learning parameterizations of atmospheric optical turbulence
  for free-space optical communication
  Current position: Postdoctoral Researcher, TU-Delft, the Netherlands
- Bowen Li (TU-Delft; co-advised with Simon Watson, Herman Russchenberg) 2025
  Dissertation: Dunkelflaute events – characterization, prediction and future projection
- Bedassa Cheneka (TU-Delft; co-advised with Simon Watson) 2023
  Dissertation: Wind power ramps – characterisation, forecasting and future projection
- Patrick Hawbecker (Atmospheric Science, NCSU) 2017
  Dissertation: The influence of ambient stability on downburst winds
  Current position: Senior Atmospheric Scientist, ArcVera Renewables
  Boulder, CO, USA

- Adam DeMarco (Atmospheric Science, NCSU; co-advisor: Russell Philbrick) 2017
  Dissertation: Multiscale characterization of the probability density functions of velocity and temperature increment fields
  Current position: Commander, 1st Combat Weather Squadron, Joint Base Lewis-McChord, USA
- Kiliyanpilakkil Velayudhan Praju (Atmospheric Science, NCSU) 2017
  Dissertation: Scaling characteristics of mesoscale wind fields in the lower atmospheric boundary layer: Implications for wind energy
- Aaron Sims (Atmospheric Science, NCSU; co-advised with Sethu Raman) 2016
  Dissertation: Investigation of the mesoscale interaction between the sea breeze circulation and the Sandhills convection
  Current position: Atmospheric Scientist, PEMDAS Technologies & Innovations, Raleigh, NC, USA
- Yao Wang (Atmospheric Science, NCSU) 2016
  Dissertation: Estimating and forecasting optical turbulence in atmosphere using an artificial neural network approach
  Current position: Principal Fundamental Analyst, NextEra Energy, Inc., West Palm Beach, FL, USA
- Chris Nunalee (Atmospheric Science, NCSU) 2015
  Dissertation: A dynamical characterization of atmospheric von Kármán vortex streets induced by bluff topography
  Current position: Director, Energy Resource Assessment and Optimization at NextEra Energy, Inc., West Palm Beach, FL, USA
- Brandon Storm (Ph.D.–IGERT Wind Science and Engineering Fellow, TTU) 2008
  Dissertation: Modeling of low-level jets over the Great Plains: Implications for Wind Energy
  Current position: Senior Meteorologist, Tradewind Energy, Inc., Lenexa, KS, USA

Former M.S. Advisees

- Kevin Schuurman (TU-Delft) 2023
  Thesis: Can Fourier Neural operators replicate the intrinsic predictability of spatiotemporal chaos for the Kuramoto-Sivashinsky system?
  Current position: Ph.D. student at TU-Delft, the Netherlands
- Gijs van Ouwerkerk (TU-Delft) 2021
  Thesis: Novel machine learning methods for short-term solar PV forecasting
- Thom Homsma (TU-Delft; co-advisor) 2021
  Thesis: Multi-step ahead ultra-short-term wind power forecasting
  Current position: Renewable energy forecast analyst at Eneco Energy Trade B.V., the Netherlands
- Juliëtte Anema (TU-Delft; co-advisor) 2021
  Thesis: An automated approach to estimate carbon monoxide emissions from steel plants by utilizing TROPOMI satellite measurements
  Current position: Ph.D. student at KNMI, the Netherlands
- Eric Lacoa Arends (TU-Delft; co-advisor) 2021
  Thesis: Novel machine learning methods to enhance wind power probabilistic forecasting
  Current position: Business Translator D&A Middle Europe at Unilever, the Netherlands

- Camilla van Wirdum (TU-Delft; co-advisor) 2020
  Thesis: Extreme convective gusts in a future warmer climate assessed through a convection permitting model
  Current position: Junior Innovation Scientist, TNO, the Netherlands
- Haolin Liu (TU-Delft) 2020
  Thesis: Characterizing coastal wind speed gradients using Scanning LiDAR data and mesoscale modeling
  Current position: Ph.D. student at Hongkong University of Science and Technology
- Qidi Yu (TU-Delft) 2020
  Thesis: Mesoscale modelling of waterspouts: an offshore wind energy perspective
  Current position: Ph.D. student at University of Bergen, Norway
- Yi Dai (TU-Delft) 2020
  Thesis: Addressing the grid-size sensitivity issue in large-eddy simulations of stable boundary layers
  Current position: Ph.D. student at TU-Delft, the Netherlands
- Kars Trommel (TU-Delft) 2020
  Thesis: Wind classification using unsupervised learning: in support of the Olympic sailing competition in Tokyo, Japan
  Current position: Management Trainee, R&D Engineering, Philips, the Netherlands
- Pooja Ramakrishna (TU-Delft; co-advised with Pier Siebesma) 2019
  Thesis: Evaluation of a wind farm parameterization in an operational mesoscale model
  Current position: InSAR Consultant, SkyGeo, Delft, the Netherlands
- Adithya Vemuri (TU-Delft; co-advised with Simon Watson) 2019
  Thesis: A new coupled modelling framework for turbine inflow generation: mesoscale-synthetic turbulence
  Current position: PhD student at von Kármán Institute for Fluid Dynamics, Belgium
- Jori Dreef (TU-Delft) 2019
  Thesis: Simulating frontal low-level jets and quantifying their impact on wind energy production
  Current position: A Sustainable Energy Advisor at Pondera Consult B.V., the Netherlands
- Sam Koch (TU-Delft) 2018
  Thesis: Characterization of inflow wind fields using SpinnerLidar measurements during ScanFlow project
  Current position: Radar meteorologist at Buienradar, the Netherlands
- Jeffrey Craft (NCSU) 2013
  Thesis: Wind ramp events: Forecast verification and climatology
  Current position: Manager, Data Analytics at Clearway Energy Group, Phoenix, AZ, USA
- Yao Wang (NCSU) 2013
  Thesis: Generating realistic inflows for utility-scale wind turbines: A large-eddy simulation-based approach
  Current position: Sr. Solutions Project Lead at NextEra Energy, Inc., West Palm Beach, FL, USA
- Heather Richardson (NCSU) 2012
  Thesis: Improving stable boundary layer parameterization in a mesoscale model to better represent nocturnal low-level jets
  Current position: Sr. Operations Engineer at WindLogics, North Palm Beach, FL, USA

- Elizabeth Wilson (NCSU, non-thesis) 2012
  Current position: Director of Weather Programs at Synoptic Data PBC, Raleigh, NC, USA
- Suraj Harshan (TTU) 2009
  Thesis: Modeling of Antarctic boundary layer
  Current position: Senior Analyst, Swiss Re, India
- Rachel Rogers (TTU) 2008
  Thesis: Forecast verification: A dispersion modeling perspective
  Current position: Wind Research Analyst II at Alliant Energy, New York, NY, USA
- Julie Phillipson (TTU) 2008
  Thesis: Bursting events in the stable atmospheric boundary layer
  Current position: National Weather Service, Midland/Odessa, TX, USA
- William Anderson (TTU) 2007
  Thesis: A localized dynamic model for large-eddy simulation of the neutrally buoyant atmospheric boundary layer
  Current position: Associate professor, Department of Mechanical Engineering, University of Texas at Dallas, TX, USA

Former Visiting Researcher

- Zihan Zhao (Ph.D. student, Harbin Institute of Technology Shenzhen, China; recipient of a CSC scholarship)
- Fabrizio Adamu (Erasmus Exchange Student from Università degli Studi di Cagliari, Italy)
- Dr. Wei Li
- Kornel Rozsavolgyi

Professional Affiliations

American Geophysical Union (AGU)
American Meteorological Society (AMS)
Directed Energy Professional Society
SPIE
The Optical Society (OSA)

SUNY SYSTEM, University, and Outreach Services

- Member, Working Group 3, STRIVE Task Force, SUNY 2026–Present
- Member, Middle States Commission on Higher Education (MSCHE), UAlbany 2026–Present
- Member, University-level T&P committee (CPCA), UAlbany 2025–Present
- Chair, Falconer seminar series, ASRC, UAlbany 2023–Present
- Member, The council on Libraries, Information Systems, and Computing (LISC), UAlbany 2024–2025
- Chair, faculty search committee, ESE Department, UAlbany 2024–2025
- Member, T&P committee, CNSE college, UAlbany 2024
- Member, President's excellence awards committee
  Excellence in librarianship, UAlbany 2023–2024
- Member, faculty search committee, ASRC, UAlbany 2024

- Member, search committee for meteorologists, SWRCC, UAlbany 2024
- Member, grant writer search committee, UAlbany 2023
- Lead coordinator, development of a new M.Sc. track on Atmospheric Environment Engineering, TU-Delft 2020–2022
- Collaboration with Sailing Innovation Centre, The Hague, The Netherlands (weather forecasting for the 2020 Olympics Games, Japan) 2018–2021
- Committee member, College of Science IT Director search, NCSU 2016
- Committee member, university-level academic misconduct committee, NCSU 2014–2015
- Committee member, college-level teaching awards committee, NCSU 2014–2015
- Science judge, Blue Heron Bowl, MEAS, NCSU 2013
- Presenter (via skype), South View Middle School, Hope Mills, NC 2012
- Presenter, Sigma-Xi 2012
- Presenter, General Electric, Greenville, SC 2012
- Interviewee, Radio In Vivo 2012
- Instructor, 'Run on the Wind: Engineering a Clean Tomorrow' summer camp, TTU 2011
- Organizer, 'Run on the Wind: Engineering a Clean Tomorrow' summer camp, TTU 2010
- Co-director, 'Run on the Wind: Engineering a Clean Tomorrow' summer camp, TTU 2009
- Chair, Numerical Weather Prediction search committee, Atmospheric Science Group, TTU 2008–2009
- Committee member, various departmental committees 2006–present